

Q1. The frames A, B are initially aligned. Frame B undergoes two consecutive rotations, first about its own x by $\frac{\pi}{2}$ and then the z axis of Frame A by $\frac{\pi}{4}$. After these two rotations:

- Find the rotation matrix of Frame B with respect to Frame A, R_B^A .
- Find the y axis of Frame A, expressed in Frame B.

Q2. The Rotation and translation of Frame B with respect to Frame A are given by:

$$R_B^A = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, O_B^A = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

A point coordinates in Frame in B is given by $p^B = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$. Find its coordinates in Frame A, $p^A = ?$.

Note that:

$$R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix}, R_y(\beta) = \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}, R_z(\gamma) = \begin{bmatrix} \cos \gamma & -\sin \gamma & 0 \\ \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Solution:

Q1)

$$a) R_B^A = R_z\left(\frac{\pi}{4}\right) R_x\left(\frac{\pi}{2}\right) = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \end{bmatrix}$$

$$b) R_A^B = (R_B^A)^T \Rightarrow y_A^B = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{bmatrix}$$

$$Q2) p^A = R_B^A p^B + O_B^A = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} + \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ -1 \end{bmatrix}$$

ELECENG 3EY4
Sample Problems on Coordinate Frames and Homogenous Transformations

- 1- Consider the following chain of rotations

$$R_4^0 = R_y(\theta_1)R_z(\theta_2)R_y(\theta_3)R_x(\theta_4).$$

Express various ways of interpreting the rotations, about fixed and current axes that would yield the same result.

- 2- Find the ZYZ Euler representation for the following rotation matrix

$$R = \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ -1 & 0 & 0 \end{bmatrix}$$

- 3- Consider the figure below. A robot is set up 1 meter from a table, two of whose legs are on the y axis as shown. The table top is 1 meter high and 1 meter square. A frame $o_1 - x_1y_1z_1$ is fixed to the edge of the table as shown. A cube measuring 20 cm on a side is placed in the center of the table with frame $o_2 - x_2y_2z_2$ established at the center of the cube as shown. A camera is situated directly above the center of the block 2 m above the tabletop with frame $o_3 - x_3y_3z_3$ attached as shown. Find the homogeneous transformations relating each of these frames to the base frame $o_0 - x_0y_0z_0$. Find the homogeneous transformation relating the camera to the object frame.

Now suppose that, after the camera is calibrated, it is rotated 90° about the axis z. Re-compute the above coordinate transformations.

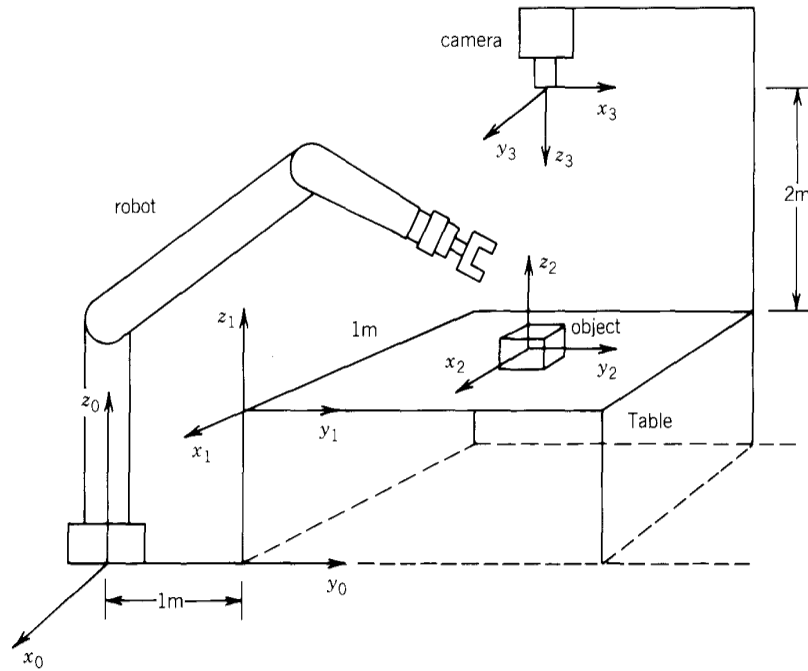


Fig. 1 (adopted from M. Spong and M. Vidyasagar, Robot Dynamics and Control, John Wiley & Sons)

- 4- In general, multiplication of homogeneous transformation matrices is not commutative. Consider the matrix product:

$$T = \text{Rot}_{x,\alpha} \text{Tran}_{x,b} \text{Tran}_{z,d} \text{Rot}_{z,\theta}$$

Here Rot and Tran are purely rotational and translational homogeneous transformations. Determine which pairs of the four matrices on the right-hand side commute. Explain why the pairs commute. Find all permutations of these four matrices that would yield the same homogeneous transformation matrix, T .

- 5- Consider the wedge-shaped object in Fig. 2. In this figure, Frame A and Frame B are attached to the rigid object whereas Frame 0 is a stationary reference frame. (Note that frames 0 & A initially coincide)

- a. Find the homogeneous transformation T_B^0 that relates Frame B to Frame 0 (0 is origin and B is destination).

- b. The coordinates of point **P** in Frame B are given by $P^B = \begin{bmatrix} 0 & 0 & 2 \end{bmatrix}^T$. Find the coordinates of this point in Frame 0, P^0 .
- c. Assume that the wedge-shaped object is moved first by translating Frame A along its x-axis by -1 units and then rotating this frame about its current y-axis by $\frac{\pi}{2}$ radians. Calculate the new coordinates of point P in Frame B, $\bar{P}^B$. (Note that point P is *not* attached to the rigid object and therefore has *not* moved itself).
- d. Assume that the movement of the object is restricted to pivoting around the origin of Frame A, i.e., the origin of the frame *cannot* move. Find a rotation for Frame A that would bring the origin of Frame B to its closest possible distance from point P. Does this problem have a unique solution? Explain.

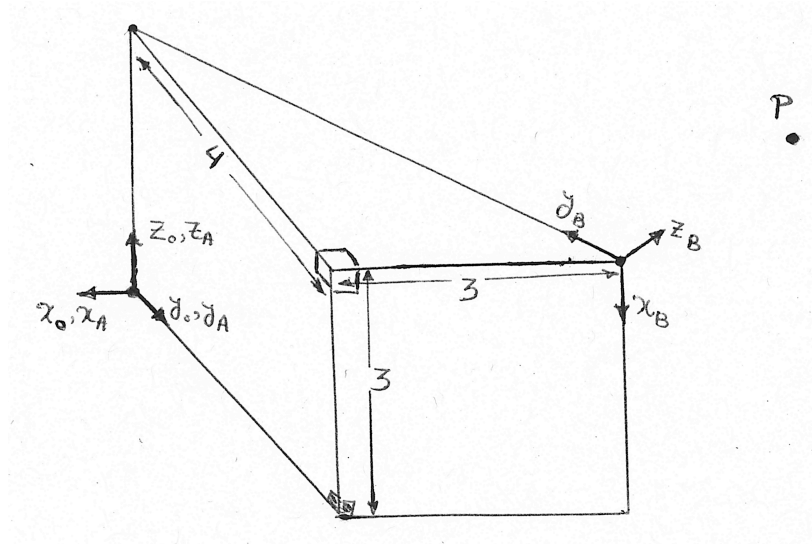


Fig. 2. The wedge-shaped object in Q5.